Year 10 Higher

REVISION PACK

Number 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 2 · Percentages (Lesson 4.1)

Q	Working	Answer
percentage loss		
1.1	$18\,000 - 14\,940 = 3060$ $\frac{3060}{18\,000} \times 100 = 17\%$	**17%**
1.2	$640 - 528 = 112$ $\frac{112}{640} \times 100 = 17.5\%$	**17.5%**
1.3	$2500 - 2150 = 350$ $\frac{350}{2500} \times 100 = 14\%$	**14%**
1.4	$12\,000 - 9000 = 3000$ $\frac{3000}{12\,000} \times 100 = 25\%$	**25%**
1.5	$450 - 396 = 54$ $\frac{54}{450} \times 100 = 12\%$	**12%**
1.6	$8000 - 7400 = 600$ $\frac{600}{8000} \times 100 = 7.5\%$	**7.5%**
reverse percentages		
1.7	$100\% - 24\% = 76\%$ $5320 \div 0.76 = 7000$	**KD 7000**
1.8	$100\% - 15\% = 85\%$ $3060 \div 0.85 = 3600$	**KD 3600**
1.9	$100\% - 8\% = 92\%$ $391 \div 0.92 = 425$	**KD 425**
1.10	$100\% - 15\% = 85\%$ $204 \div 0.85 = 240$	**KD 240**

1.11	$100\% - 5\% = 95\%$ $2850 \div 0.95 = 3000$	**KD 3000**
1.12	$100\% - 20\% = 80\%$ $8320 \div 0.8 = 10\,400$	**KD 10 400**

Number 2 · Recurring decimals (Lesson 4.2)

Q	Working	Answer
recurring decimal proofs		
2.1	$x = 0.27777777\ldots$ $100x = 27.777777\ldots$ $10x = 2.777777\ldots$ $90x = 25$ $x = \frac{25}{90} = \frac{5}{18}$	$x = \frac{5}{18}$ **shown**
2.2	$x = 0.38888888\ldots$ $100x = 38.888888\ldots$ $10x = 3.888888\ldots$ $90x = 35$ $x = \frac{35}{90} = \frac{7}{18}$	$x = \frac{7}{18}$ **shown**
2.3	$x = 0.41111111\ldots$ $100x = 41.111111\ldots$ $10x = 4.111111\ldots$ $90x = 37$ $x = \frac{37}{90} = \frac{37}{90}$	$x = \frac{37}{90}$ **shown**
2.4	$x = 0.15555555\ldots$ $100x = 15.555555\ldots$ $10x = 1.555555\ldots$ $90x = 14$ $x = \frac{14}{90} = \frac{7}{45}$	$x = \frac{7}{45}$ **shown**

2.5	$x = 0.25555555\ldots$ $100x = 25.555555\ldots$ $10x = 2.555555\ldots$ $90x = 23$ $x = \frac{23}{90} = \frac{23}{90}$	$x = \frac{23}{90}$ **shown**
2.6	$x = 0.52222222\ldots$ $100x = 52.222222\ldots$ $10x = 5.222222\ldots$ $90x = 47$ $x = \frac{47}{90} = \frac{47}{90}$	$x = \frac{47}{90}$ **shown**

Number 2 · Percentages (Lesson 4.1)

Q	Working	Answer
increasing and decreasing by a percentage		
3.1	$0.85 \times 480 = 408$	**KD 408**
3.2	$0.78 \times 1250 = 975$	**KD 975**
3.3	$0.65 \times 86 = 55.9$	**KD 55.90**
3.4	$0.925 \times 2400 = 2220$	**KD 2220**
3.5	$1.18 \times 340 = 401.2$	**KD 401.20**
3.6	$1.032 \times 15000 = 15480$	**KD 15 480**

Number 2 · Percentages (Lesson 4.1)

Q	Working	Answer
percentage increase		
3.7	$33440 - 32000 = 1440$ $\frac{1440}{32000} \times 100$	**4.5%**
3.8	$285 - 250 = 35$ $\frac{35}{250} \times 100$	**14%**
3.9	$1953 - 1800 = 153$ $\frac{153}{1800} \times 100$	**8.5%**
3.10	$80 - 64 = 16$ $\frac{16}{64} \times 100$	**25%**
3.11	$13250 - 12500 = 750$ $\frac{750}{12500} \times 100$	**6%**
3.12	$93 - 75 = 18$ $\frac{18}{75} \times 100$	**24%**